



Software Product Developer

QP Code: SSC/Q0512

Version: 1.0

NSQF Level: 5

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Qualification Pack

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SSC/Q0512: Software Product Developer

Brief Job Description

A Software Product Developer uses specialized knowledge of programming languages and software integration to build products and deliver business solutions. Their responsibilities include application enhancement, debugging, maintenance, and documentation. They handle the full development lifecycle, from coding to deployment, while collaborating with teams to integrate new features and ensure functionality. Additionally, they stay updated on industry trends and technologies

Personal Attributes

The individual should be self-motivated, able to work independently, and make informed decisions related to their work. They should be result-oriented, with strong communication skills and the ability to think logically to solve problems effectively.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N0514: Develop and Design Software Products and Applications](#)
2. [SSC/N0502: Develop Software Code Based on Specifications](#)
3. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Software Product Development
Occupation	Application Deployment
Country	India
NSQF Level	5
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2511.0103

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Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (of 3-year/ 4-year UG. In Science, Technology, Engineering, And Mathematics (STEM)) OR Completed 3-year diploma (after 10th) with 1.5 years of experience in relevant field.(Relevant Experience in Software Development The relevant experience would include work, internship and apprenticeship after completion of relevant educational qualification.) OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience in relevant field.(Relevant Experience in Software Development The relevant experience would include work, internship and apprenticeship after completion of relevant educational qualification.)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	1.0
Reference code on NQR	QG-05-IT-03651-2025-V2-NASSCOM
NQR Version	2

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SSC/N0514: Develop and Design Software Products and Applications

Description

This NOS focuses on the competencies required for designing, developing, and deploying software products and applications. It includes tasks such as understanding user requirements, applying UI/UX principles, coding, testing, debugging, and collaborating with cross-functional teams to deliver robust and scalable solutions. It emphasizes the incorporation of emerging technologies and best practices to meet industry standards

Scope

The scope covers the following :

- Understand Requirements and Define Architecture
- Incorporate UI/UX Principles in Software Design
- Develop and Implement Software Solutions
- Perform Testing, Debugging, and Optimization
- Maintain and Document Software Applications
- Collaborate with Teams for Development and Deployment
- Leverage Gen AI for Software Design and Development

Elements and Performance Criteria

Understand Requirements and Define Architecture

To be competent, the user/individual on the job must be able to:

- PC1.** analyze user requirements and create detailed technical specifications.
- PC2.** define software architecture and system design based on functional and non-functional requirements.
- PC3.** evaluate feasibility and risks, ensuring scalability and performance optimization.

Incorporate UI/UX Principles in Software Design

To be competent, the user/individual on the job must be able to:

- PC4.** understand and apply UI/UX principles to ensure user-friendly interfaces and workflows
- PC5.** collaborate with UI/UX designers to translate wireframes and mockups into functional interfaces.
- PC6.** conduct usability testing and incorporate user feedback into design refinements

Develop and Implement Software Solutions

To be competent, the user/individual on the job must be able to:

- PC7.** write clean, maintainable, and scalable code in accordance with coding standards
- PC8.** implement features and functionalities aligned with requirements and system architecture.
- PC9.** integrate third-party APIs and libraries as required by the project.

Perform Testing, Debugging, and Optimization

To be competent, the user/individual on the job must be able to:

- PC10.** develop and execute unit tests to validate code functionality

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PC11. identify and resolve defects and performance issues through debugging and profiling tools.

PC12. optimize code and database queries for enhanced performance and efficiency

Maintain and Document Software Applications

To be competent, the user/individual on the job must be able to:

PC13. document software architecture, modules, and changes for future reference

PC14. maintain version control and update software as required

PC15. provide support for software deployment and post-launch maintenance.

Collaborate with Teams for Development and Deployment

To be competent, the user/individual on the job must be able to:

PC16. work with cross-functional teams, including business analysts, QA testers, and operations teams

PC17. participate in Agile or Scrum meetings to plan, prioritize, and review progress.

PC18. ensure successful integration and deployment in production environments

Leverage Gen AI for Software Design and Development

To be competent, the user/individual on the job must be able to:

PC19. understand the fundamentals of Generative AI (Gen AI), including its capabilities, limitations, and ethical considerations

PC20. explore Gen AI use cases, such as code generation, automated testing, and prototyping

PC21. apply prompt engineering techniques to leverage Gen AI tools for designing, developing, and optimizing software applications.

PC22. integrate AI-powered platforms to automate repetitive tasks and improve efficiency during software development.

PC23. evaluate the performance and applicability of AI-generated solutions and incorporate improvements.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the software development life cycle, including Agile and Waterfall methodologies

KU2. principles of UI/UX design and their application in software development.

KU3. system architecture and design patterns.

KU4. programming languages, frameworks, and libraries used in software development.

KU5. testing and debugging techniques, including unit, integration, and system testing

KU6. tools for version control (e.g., Git) and collaborative development

KU7. emerging technologies such as AI, cloud computing, and IoT

KU8. best practices for code documentation and maintenance

KU9. the strategies for optimizing performance, scalability, and security in software products.

KU10. techniques for seamless integration and deployment of software applications

KU11. the basics of Generative AI, including its applications in software development and testing.

KU12. the methods for creating and refining AI prompts to achieve desired outputs.

KU13. ethical considerations and best practices for integrating Gen AI in software projects

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KU14. industry-standard AI tools and platforms for enhancing software development processes.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to learn about the latest developments in the work
- GS3.** listen attentively to understand the information/ instructions being shared by the speaker
- GS4.** communicate clearly and politely with co-workers and clients
- GS5.** coordinate with co-workers to achieve work objectives
- GS6.** plan and prioritize tasks to ensure timely completion
- GS7.** identify possible disruptions to work and take appropriate preventive measures
- GS8.** take quick decisions to deal with workplace emergencies/ accidents
- GS9.** evaluate all possible solutions to a problem to select the best one

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand Requirements and Define Architecture</i>	4	6	-	3
PC1. analyze user requirements and create detailed technical specifications.	1	2	-	1
PC2. define software architecture and system design based on functional and non-functional requirements.	2	2	-	1
PC3. evaluate feasibility and risks, ensuring scalability and performance optimization.	1	2	-	1
<i>Incorporate UI/UX Principles in Software Design</i>	6	9	-	3
PC4. understand and apply UI/UX principles to ensure user-friendly interfaces and workflows	2	3	-	1
PC5. collaborate with UI/UX designers to translate wireframes and mockups into functional interfaces.	2	3	-	1
PC6. conduct usability testing and incorporate user feedback into design refinements	2	3	-	1
<i>Develop and Implement Software Solutions</i>	6	10	-	3
PC7. write clean, maintainable, and scalable code in accordance with coding standards	2	3	-	1
PC8. implement features and functionalities aligned with requirements and system architecture.	2	4	-	1
PC9. integrate third-party APIs and libraries as required by the project.	2	3	-	1
<i>Perform Testing, Debugging, and Optimization</i>	3	5	-	3
PC10. develop and execute unit tests to validate code functionality	1	2	-	1
PC11. identify and resolve defects and performance issues through debugging and profiling tools.	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. optimize code and database queries for enhanced performance and efficiency	1	1	-	1
<i>Maintain and Document Software Applications</i>	3	4	-	2
PC13. document software architecture, modules, and changes for future reference	1	1	-	1
PC14. maintain version control and update software as required	1	2	-	0.5
PC15. provide support for software deployment and post-launch maintenance.	1	1	-	0.5
<i>Collaborate with Teams for Development and Deployment</i>	3	6	-	3
PC16. work with cross-functional teams, including business analysts, QA testers, and operations teams	1	2	-	1
PC17. participate in Agile or Scrum meetings to plan, prioritize, and review progress.	1	2	-	1
PC18. ensure successful integration and deployment in production environments	1	2	-	1
<i>Leverage Gen AI for Software Design and Development</i>	5	10	-	3
PC19. understand the fundamentals of Generative AI (Gen AI), including its capabilities, limitations, and ethical considerations	1	2	-	1
PC20. explore Gen AI use cases, such as code generation, automated testing, and prototyping	1	2	-	0.5
PC21. apply prompt engineering techniques to leverage Gen AI tools for designing, developing, and optimizing software applications.	1	2	-	0.5
PC22. integrate AI-powered platforms to automate repetitive tasks and improve efficiency during software development.	1	2	-	0.5
PC23. evaluate the performance and applicability of AI-generated solutions and incorporate improvements.	1	2	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0514
NOS Name	Develop and Design Software Products and Applications
Sector	IT-ITeS
Sub-Sector	
Occupation	Application Deployment
NSQF Level	5
Credits	6
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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SSC/N0502: Develop Software Code Based on Specifications

Description

This NOS focuses on the competency to develop software code that aligns with specified requirements, ensuring that the code is efficient, maintainable, and meets performance standards. It also emphasizes the use of Gen AI tools and techniques to enhance coding practices and execution.

Scope

The scope covers the following :

- Interpret Software Specifications and Design Code
- Write Efficient and Maintainable Code
- Follow Coding Standards and Guidelines
- Collaborate with Team Members to Review Code
- Utilize Tools and Platforms for Code Development
- Monitor and Maintain Software Code Post-Development
- Apply Prompt Engineering for Software Code Enhancement
- Unit Testing and Automation

Elements and Performance Criteria

Interpret Software Specifications and Design Code

To be competent, the user/individual on the job must be able to:

- PC1.** interpret software specifications accurately to understand the required functionality
- PC2.** analyze and break down technical requirements into coding tasks.
- PC3.** Access reusable components, code generation tools, and unit testing tools from the organization's knowledge base
- PC4.** ensure alignment of the code with project requirements and design guidelines.
- PC5.** identify potential issues in the specifications and clarify with stakeholders
- PC6.** maintain proper documentation for high-level and low-level design.

Write Efficient and Maintainable Code

To be competent, the user/individual on the job must be able to:

- PC7.** develop modular, efficient, and maintainable software code, ensuring readability and scalability.
- PC8.** test, debug, and optimize code to meet intended functionality, performance, and scalability requirements.

Follow Coding Standards and Guidelines

To be competent, the user/individual on the job must be able to:

- PC9.** follow industry-standard coding practices and organizational guidelines to ensure consistency and quality
- PC10.** maintain consistent code formatting, naming conventions, and documentation for clarity and adherence to best practices.

Collaborate with Team Members to Review Code

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To be competent, the user/individual on the job must be able to:

- PC11.** collaborate with team members for code reviews, incorporating feedback to improve code quality and resolve issues.
- PC12.** use version control tools like Git to ensure effective collaboration and maintain code specifications

Utilize Tools and Platforms for Code Development

To be competent, the user/individual on the job must be able to:

- PC13.** use integrated development environments (IDEs) and other tools for coding.
- PC14.** manage version control and handle merges effectively.
- PC15.** leverage automated testing tools to validate the correctness of the code

Monitor and Maintain Software Code Post-Development

To be competent, the user/individual on the job must be able to:

- PC16.** monitor and optimize software performance post-deployment, addressing issues and implementing updates based on user feedback and performance metrics.
- PC17.** debug and resolve errors, ensuring deployment-ready code with proper documentation and reviews
- PC18.** conduct peer reviews, incorporate feedback, and finalize code before handoff.

Apply Prompt Engineering for Software Code Enhancement

To be competent, the user/individual on the job must be able to:

- PC19.** use prompt engineering techniques to generate better code suggestions with Gen AI tools.
- PC20.** apply prompt engineering in testing scenarios to create efficient test cases

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the software development life cycle (SDLC) and its stages, including requirements gathering, design, coding, testing, deployment, and maintenance.
- KU2.** the various software design techniques, such as object-oriented design, modular programming, and design patterns
- KU3.** how to interpret software specifications to ensure that the developed code meets the functional and non-functional requirements
- KU4.** how to write efficient, maintainable, and reusable code to ensure long-term code health and ease of modification
- KU5.** the version control systems like Git and the importance of managing code versions in a collaborative environment
- KU6.** the coding best practices and standards, including naming conventions, commenting, error handling, and ensuring code readability
- KU7.** the debugging techniques and tools to identify, fix, and optimize errors in the code
- KU8.** the performance optimization strategies, such as code refactoring, caching, and database query optimization
- KU9.** the Gen AI tools and how they can be leveraged to automate repetitive coding tasks, optimize code, and generate test cases

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- KU10.** the importance of maintaining high-quality code and how prompt engineering can help in generating better code suggestions
- KU11.** the unit testing frameworks and automated testing tools used in the software development process
- KU12.** the significance of peer reviews and how feedback can improve code quality and adherence to specifications
- KU13.** the tools and platforms that support collaborative development, code versioning, and deployment readiness
- KU14.** the deployment processes and the steps required to ensure the code is ready for deployment
- KU15.** the code handoff procedures, including proper documentation and communication with stakeholders
- KU16.** the modularization and reusability principles for designing software components that can be reused in different parts of the application or across projects.
- KU17.** the AI and prompt engineering techniques to optimize code generation, enhance debugging processes, and improve the quality of software development.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to learn about the latest developments in the work
- GS3.** listen attentively to understand the information/ instructions being shared by the speaker
- GS4.** communicate clearly and politely with co-workers and clients
- GS5.** coordinate with co-workers to achieve work objectives
- GS6.** plan and prioritize tasks to ensure timely completion
- GS7.** identify possible disruptions to work and take appropriate preventive measures
- GS8.** take quick decisions to deal with workplace emergencies/ accidents
- GS9.** evaluate all possible solutions to a problem to select the best one

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Interpret Software Specifications and Design Code</i>	7	15	-	6
PC1. interpret software specifications accurately to understand the required functionality	1	3	-	1
PC2. analyze and break down technical requirements into coding tasks.	2	3	-	1
PC3. Access reusable components, code generation tools, and unit testing tools from the organization's knowledge base	1	3	-	1
PC4. ensure alignment of the code with project requirements and design guidelines.	1	2	-	1
PC5. identify potential issues in the specifications and clarify with stakeholders	1	2	-	1
PC6. maintain proper documentation for high-level and low-level design.	1	2	-	1
<i>Write Efficient and Maintainable Code</i>	4	6	-	2
PC7. develop modular, efficient, and maintainable software code, ensuring readability and scalability.	2	3	-	1
PC8. test, debug, and optimize code to meet intended functionality, performance, and scalability requirements.	2	3	-	1
<i>Follow Coding Standards and Guidelines</i>	2	6	-	2
PC9. follow industry-standard coding practices and organizational guidelines to ensure consistency and quality	1	3	-	1
PC10. maintain consistent code formatting, naming conventions, and documentation for clarity and adherence to best practices.	1	3	-	1
<i>Collaborate with Team Members to Review Code</i>	3	6	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. collaborate with team members for code reviews, incorporating feedback to improve code quality and resolve issues.	2	3	-	1
PC12. use version control tools like Git to ensure effective collaboration and maintain code specifications	1	3	-	1
<i>Utilize Tools and Platforms for Code Development</i>	5	6	-	3
PC13. use integrated development environments (IDEs) and other tools for coding.	2	2	-	1
PC14. manage version control and handle merges effectively.	2	2	-	1
PC15. leverage automated testing tools to validate the correctness of the code	1	2	-	1
<i>Monitor and Maintain Software Code Post-Development</i>	5	6	-	3
PC16. monitor and optimize software performance post-deployment, addressing issues and implementing updates based on user feedback and performance metrics.	2	2	-	1
PC17. debug and resolve errors, ensuring deployment-ready code with proper documentation and reviews	2	2	-	1
PC18. conduct peer reviews, incorporate feedback, and finalize code before handoff.	1	2	-	1
<i>Apply Prompt Engineering for Software Code Enhancement</i>	4	5	-	2
PC19. use prompt engineering techniques to generate better code suggestions with Gen AI tools.	2	3	-	1
PC20. apply prompt engineering in testing scenarios to create efficient test cases	2	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0502
NOS Name	Develop Software Code Based on Specifications
Sector	IT-ITeS
Sub-Sector	IT Services
Occupation	Application Deployment
NSQF Level	5
Credits	8
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0514.Develop and Design Software Products and Applications	30	50	-	20	100	45
SSC/N0502.Develop Software Code Based on Specifications	30	50	-	20	100	45
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	80	130	-	40	250	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
IT-ITeS	Information Technology - Information Technology enabled Services
BPM	Business Process Management
BPO	Business Process Outsourcing
KPO	Knowledge Process Outsourcing
LPO	Legal Process Outsourcing

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Helpdesk	Helpdesk is an entity to which the customers will report their IT problems. IT Service Helpdesk Attendant is responsible for managing the helpdesk.